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Course Code : CSE 260

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Section: 27A

### [Assignment - 2]

[Ans to the question no 1]

$$a) (P+Q+R)(P+Q')(P'+R)$$

$$= (P+Q+R)(PP' + PR + Q'P + Q'R)$$

$$= (P+Q+R)(PR + Q'P + Q'R)$$

$$= (P \cdot PR + P \cdot Q'P + P \cdot Q'R + Q \cdot PR + Q \cdot Q'P + Q \cdot Q'R + R \cdot PR + R \cdot Q'P + R \cdot Q'R)$$

$$= (PR + PQ'R + PQR + PR + P'Q'R + Q'R)$$

$$= (PR + PQR + PQ'R + P'Q'R + Q'R)$$

$$= PR(1+Q+Q') + P'Q'R + Q'R$$

$$= PR(1) + P'Q'R + Q'R$$

$$= PR + P'Q'R + Q'R$$

$$= R(P + P'Q' + Q')$$

$$= R(P + Q'(P'+1))$$

$$= R(P + Q')$$

$$= RP + Q'R$$

(Ans)

$$b) A'B'c + (A+B+C)' + A'B'c'$$

$$= A'B'(c+c') + (A+B+C)'$$

$$= A'B' + (A' \cdot B' \cdot c')$$

$$= A'B'(1+c')$$

$$= A'B'$$

(Ans)

$$c) ((AB)'. (C+D)')' + (A \oplus (B+C))$$

$$= (A'+B')' + (C+D) + A(B'C') + A'(B+C)$$

$$= AB + C + D + AB'C' + A'B + A'C$$

$$= B(A+A') + C + D + AB'C' + A'C$$

$$= B + C + D + A'C + AB'C'$$

$$= B + C(1+A') + D + AB'C'$$

$$= B + C + D + AB'C'$$

$$= B + D + (C + AB'C')$$

$$= B + D + (C + C')(C + AB')$$

$$= B + D + C + AB'$$

$$= C + D + (B + AB')$$

$$= C + D + (B + B')(B + A)$$

$$= C + D + B + A$$

$$= A + B + C + D$$

(Ans)

[Ans to the question no 2]

$$A' + C(B+C')(A+B')(A+C')$$

$$E_{\text{dual}} = A'(C + BC' + A'B' + AC')$$

$$E'_{\text{dual}} = A(C' + B'C + AB + A'C)$$

$$= AC' + AB'C + AB + A'C \cdot A$$

$$= AC' + AB'C + AB$$

$$= A(C' + B'C + B)$$

$$= A(C' + B' + B)$$

$$= A(C' + 1)$$

$$= A$$



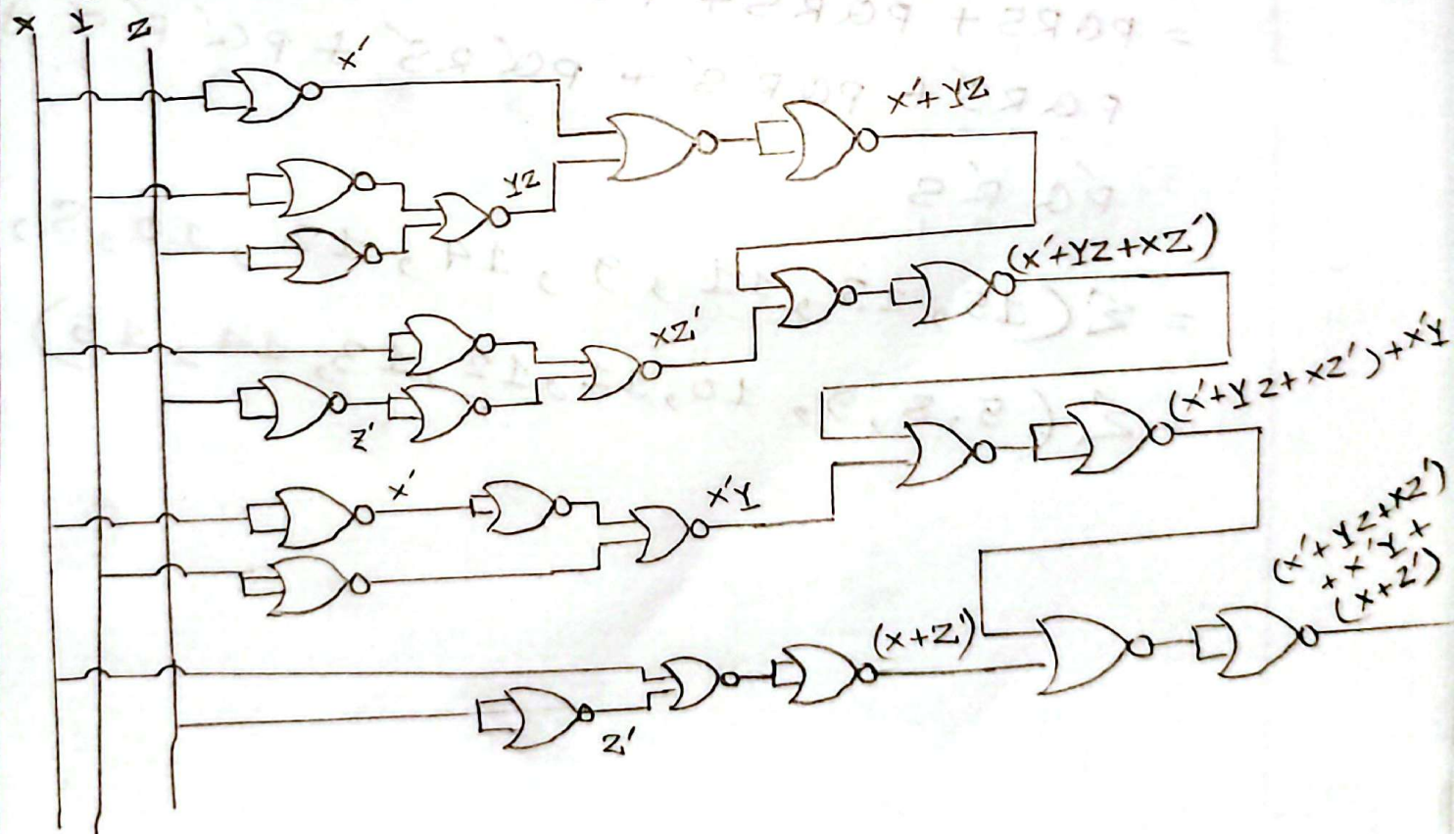
[Ans to the question no 3]

$$\pi(0, 2, 7)$$

$$F(A, B, C) = \text{ABC} (A+B+C)(A+B'+C)(A'+B+C)$$

[Ans to the question no 4]

$$F(x, y, z) = (x' + yz + xz') + x'y + (x + z')$$



[Ans to the question no 5]

SoP  
x=1  
x'=0

$$F(P, Q, R, S) = P + QR'S$$

$$= P(Q+Q')(R+R')(S+S') + (P+P')QR'S$$

$$= (PQ+PQ')(R+R')(S+S') + (PQR'S) + (P'QR'S)$$

$$= (PQR + PQR' + PQR + PQR') (S+S')$$

$$+ PQR'S + P'QR'S$$

$$= PQR S + PQR'S + PQR'S + PQR'S +$$

$$PQR'S + PQR'S + PQR'S + PQR'S +$$

$$PQR'S + P'QR'S$$

$$= \underset{0}{P} \underset{0}{Q} \underset{1}{R} \underset{1}{S} + \underset{0}{P} \underset{0}{Q} \underset{1}{R} \underset{0}{S} + \underset{0}{P} \underset{1}{Q} \underset{1}{R} \underset{1}{S} + \underset{0}{P} \underset{1}{Q} \underset{0}{R} \underset{1}{S} +$$

$$\underset{1}{P} \underset{1}{Q} \underset{1}{R} \underset{0}{S} + \underset{1}{P} \underset{1}{Q} \underset{0}{R} \underset{0}{S} + \underset{1}{P} \underset{0}{Q} \underset{1}{R} \underset{0}{S} + \underset{1}{P} \underset{0}{Q} \underset{0}{R} \underset{0}{S} +$$

$$\underset{0}{P} \underset{1}{Q} \underset{0}{R} \underset{1}{S}$$

$$= \sum (15, 13, 11, 9, 14, 12, 10, 8, 5)$$

$$= \sum (5, 8, 9, 10, 11, 12, 13, 14, 15)$$